Tarea M21-CD - YAIR PALOMARES

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import os
os.chdir("C:/Users/yair_/OneDrive/Desktop/Ciencia de Datos/Módulo 21 - KNN y métricas de clasificación")
```

```
data = pd.read_csv("recursos_humanos.csv")
data
```

	satisfaction_level	last_evaluation	number_project	average_montly_hours	time_spend_company	Work_accident	left	promotion_last_5years	sales	salary
0	0.38	0.53	2	157	3	0	1	0	sales	low
1	0.80	0.86	5	262	6	0	1	0	sales	medium
2	0.11	0.88	7	272	4	0	1	0	sales	medium
3	0.72	0.87	5	223	5	0	1	0	sales	low
4	0.37	0.52	2	159	3	0	1	0	sales	low
...	...	...	...	...	...	...	...	...	...	...
14994	0.40	0.57	2	151	3	0	1	0	support	low
14995	0.37	0.48	2	160	3	0	1	0	support	low
14996	0.37	0.53	2	143	3	0	1	0	support	low
14997	0.11	0.96	6	280	4	0	1	0	support	low
14998	0.37	0.52	2	158	3	0	1	0	support	low

14999 rows × 10 columns

```python
#Normalización de columnas faltantes
data["average_montly_hours"] = (data["average_montly_hours"] - np.min(data["average_montly_hours"])) / (np.max(data["average_montly_hours"]) - np.min(da
data["time_spend_company"] = (data["time_spend_company"] - np.min(data["time_spend_company"])) / (np.max(data["time_spend_company"]) - np.min(data["time_

#Se crean las variables dummy para las variables categóricas
number_project_dummy = pd.get_dummies(data["number_project"], prefix="Project", dtype= np.uint8)
number_sales_dummy = pd.get_dummies(data["sales"], prefix="Sales", dtype = np.uint8)
number_salary_dummy = pd.get_dummies(data["salary"], prefix="Salary", dtype = np.uint8)

#Se concatenan las columnas de las variables dummy creadas
data = pd.concat([data, number_project_dummy], axis=1)
data = pd.concat([data, number_sales_dummy], axis=1)
data = pd.concat([data, number_salary_dummy], axis=1)
data
```

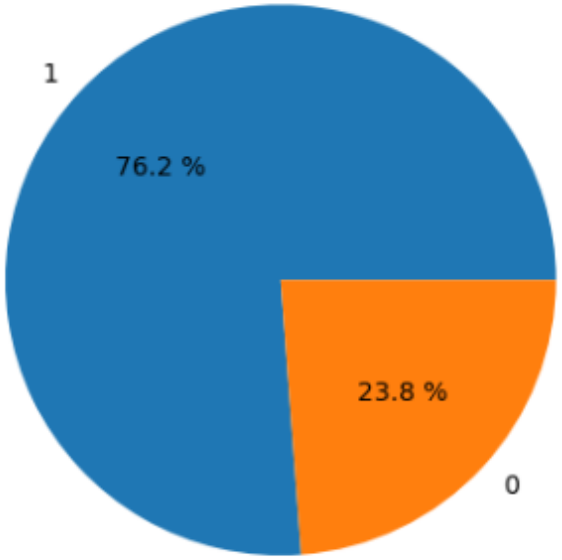
	satisfaction_level	last_evaluation	number_project	average_montly_hours	time_spend_company	Work_accident	left	promotion_last_5years	sales	salary	...
0	0.38	0.53	2	0.285047	0.125	0	1	0	sales	low	...
1	0.80	0.86	5	0.775701	0.500	0	1	0	sales	medium	...
2	0.11	0.88	7	0.822430	0.250	0	1	0	sales	medium	...
3	0.72	0.87	5	0.593458	0.375	0	1	0	sales	low	...
4	0.37	0.52	2	0.294393	0.125	0	1	0	sales	low	...
...	...	...	...	...	...	...	...	...	...	...	...
14994	0.40	0.57	2	0.257009	0.125	0	1	0	support	low	...
14995	0.37	0.48	2	0.299065	0.125	0	1	0	support	low	...
14996	0.37	0.53	2	0.219626	0.125	0	1	0	support	low	...
14997	0.11	0.96	6	0.859813	0.250	0	1	0	support	low	...
14998	0.37	0.52	2	0.289720	0.125	0	1	0	support	low	...

14999 rows × 29 columns

```
#Análisis exploratorio de los datos para validar equilibrio
frec_left = pd.value_counts(data["left"])
etiquetas = list(data['left'].unique())
plt.pie(frec_left, labels=etiquetas , autopct="%0.1f %%")
plt.show
```

```
C:\Users\yair_\AppData\Local\Temp\ipykernel_18856\3730828639.py:2: FutureWarning: pandas.value_counts is deprecated and will be removed in a future vers
ion. Use pd.Series(obj).value_counts() instead.
  frec_left = pd.value_counts(data["left"])
```

```
<function matplotlib.pyplot.show(close=None, block=None)>
```




```
#Se eliminan las columnas de las variables categóricas
data.drop(["number_project", "sales", "salary"], axis=1, inplace=True)
```

```python
#Se definen variables dependientes e independientes
y = data.left.values
x = data.drop(["left"], axis=1)

#Base de entrenamiento y prueba
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x,y,test_size=0.3, random_state=1)

#Modelo KNN (K vecinos más cercanos)
from sklearn.neighbors import KNeighborsClassifier

scores = {}
for i in range(1,21):
    knn = KNeighborsClassifier(n_neighbors=i)
    knn.fit(x_train, y_train)
    prediction = knn.predict(x_test)
    scores[str(i)] = knn.score(x_test, y_test)
```

```python
#Tabla de evaluación de 1-20 vecinos
evaluation_table = pd.DataFrame([scores]).transpose()
evaluation_table.columns = ["Evaluación"]
evaluation_table
```

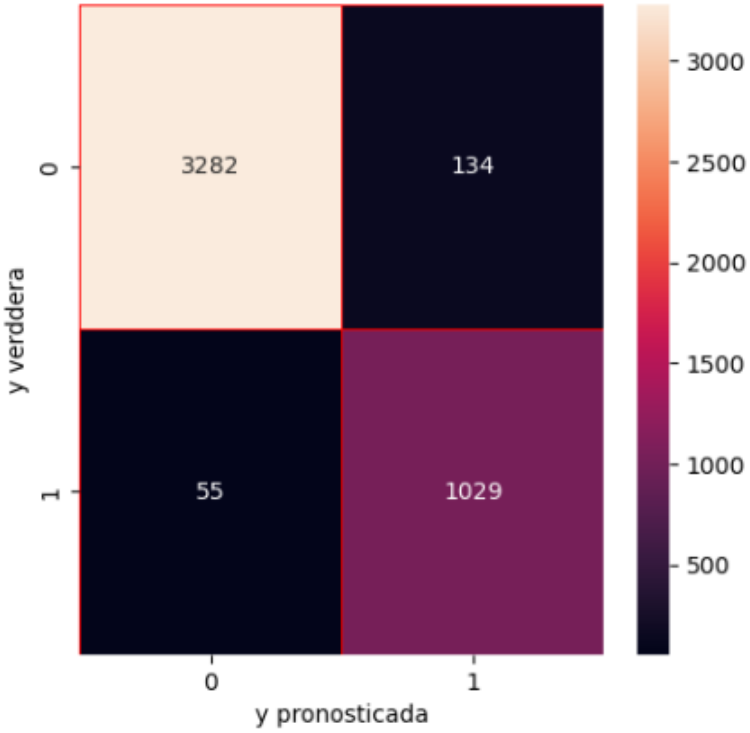
	Evaluación
1	0.958000
2	0.954667
3	0.939333
4	0.942667
5	0.936667
6	0.940222
7	0.931556
8	0.930667
9	0.927111
10	0.927778
11	0.921556
12	0.922444
13	0.917556
14	0.917111
15	0.914000
16	0.916000
17	0.911333
18	0.913556
19	0.909333
20	0.908889

```python
#Modelo KNN con el valor óptimo de K=1
knn = KNeighborsClassifier(n_neighbors=1)
knn.fit(x_train, y_train)
prediction = knn.predict(x_test)
scores[str(i)] = knn.score(x_test, y_test)

y_pred = prediction
y_true = y_test

#Creación de la matriz de confusión
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_true, y_pred)

#Visualización de la matriz de confusión
import seaborn as sns
f, ax = plt.subplots(figsize = (5, 5))
sns. heatmap(cm, annot=True, linewidths=0.5, linecolor="red", fmt=".0f", ax=ax)
plt.xlabel("y pronosticada")
plt.ylabel("y verddera")
plt.show()
```



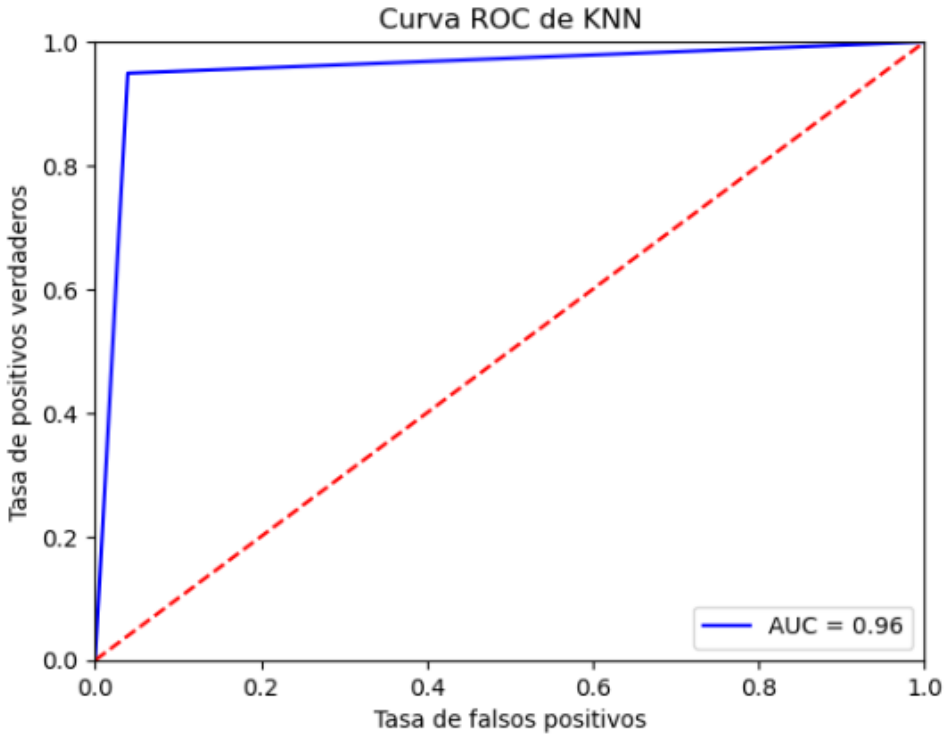
Cuando el algoritmo pronosticó que el empleado no renuncia, acertó en 3,282 ocasiones, mientras que falló en 55. Por otro lado, cuando el algoritmo pronosticó que el empleado renunciaba, logró acertar en 1,029 ocasiones y fallar en 134.

```
#Cálculo de la precisión global
correctos = cm[0,0] + cm[1,1]
incorrectos = cm[0,1] + cm[1,0]
PrecisionGlobal = correctos / (correctos + incorrectos)

#Precisión de que no renuncie
Precision_NoRenuncia = cm[0,0] / (cm[0,0] + cm[1,0])

#Precisión de que renuncie
Precision_Renuncia = cm[1,1] / (cm[0,1] + cm[1,1])

print("Precisión Global del Algoritmo = ", PrecisionGlobal)
print("Precisión para predecir cuando un empleado No Renuncia = ", Precision_NoRenuncia)
print("Precisión para predecir cuando un empleado Renuncia = ", Precision_Renuncia)
```

```
Precisión Global del Algoritmo =  0.958
Precisión para predecir cuando un empleado No Renuncia =  0.9835181300569373
Precisión para predecir cuando un empleado Renuncia =  0.884780739466896
```

```python
from sklearn.metrics import roc_curve
from sklearn.metrics import auc

y_scores = knn.predict_proba(x_test)

fpr, tpr, threshold = roc_curve(y_test, y_scores[:,1])
# fpr = False Positive Rate
# tpr = True Positive Rate
roc_auc = auc(fpr, tpr)

plt.title("Receiver Operating Characteristic")
plt.plot(fpr, tpr, "b", label="AUC = %0.2f" % roc_auc)
plt.legend(loc = "lower right")
plt.plot([0,1], [0,1], "r--")
plt.xlim([0,1])
plt.ylim([0,1])
plt.ylabel("Tasa de positivos verdaderos")
plt.xlabel("Tasa de falsos positivos")
plt.title("Curva ROC de KNN")
plt.show()
```